Wina Bwangu Web App — Project Breakdown

Due: 17 September 2026 | Total: 50 marks | Scope: Client-side prototype only (HTML/CSS/JS)

The gist

We're building a browser-based prototype for a mobile-money company (Wina Bwangu) with 6 booths offering 5 different services (Airtel Money, MTN, Zamtel, Zanaco, FNB), each with its own revenue rate and monthly transaction limit. Since this is the **client-side** exercise, there's no real server/database — all 308 transactions live in JS data (array/JSON) and everything runs in the browser.

Core flow: pick Booth → Location auto-fills → pick Service (only ones that booth actually offers) → Revenue Rate auto-fills → system generates a TransactionID → tax is calculated → dashboard updates.

Grading components (what marks are tied to)

- Booth/Service dropdowns + auto-fill logic
- TransactionID auto-generation
- Tax calculation + data bars showing tax performance
- All 308 transactions pre-loaded from Appendix 1
- Dashboard: cumulative totals & remaining monthly limit per service
- Dashboard: cumulative revenue per booth
- Dashboard: service frequency per booth
- Dashboard: total revenue + total capital summary (2 marks)
- Pie chart of totals (1 mark)
- 1000-word report (20 marks) — biggest single chunk
- Fluid navigation, form validation, alert/toast feedback (logbook requirement)

Team Assignment (7 people)

1. Project Lead / Integration Manager

- Owns the repo/folder structure and final assembly of everyone's pieces into one working prototype
- Builds the navigation shell (menu, page/section switching)
- Runs final integration testing before submission
- Keeps everyone unblocked and on schedule

2. Data & Logic Engineer

- Builds the core JS data model: booths → locations, booths → offered services, service → revenue rate, service → monthly limit
- Writes the calculation functions: transaction amount after tax, cumulative totals, revenue per booth, remaining monthly limit
- **This unblocks almost everyone else — start first**

3. Data Population Specialist

- Transcribes all 308 transactions from Appendix 1 into the JS data array/JSON used by the app
- Double-checks entries against the table (booth, location, service, rate, amount) for typos/mismatches
- **Also start immediately** — tedious and independent of other work, but the dashboard/charts can't be tested without it

4. Forms & Input UI Developer

- Builds the transaction entry form: Booth dropdown → Location auto-display, Service dropdown (filtered to that booth) → Revenue Rate auto-display
- Implements TransactionID auto-generation on form submit
- Depends on #2's data model for the dropdown logic

5. Validation & Feedback Developer

- Adds JS/Ajax-style validation across all forms (required fields, valid selections, no duplicate/invalid transactions)

- Implements alert/toast notifications for success, errors, and confirmations
- Works closely with #4 since validation sits on top of the form UI

6. Dashboard Developer

- Builds the dashboard: cumulative totals & amount remaining per service, cumulative revenue per booth, service frequency per booth, total revenue, total capital
- Implements the data bars showing tax obligation performance
- Depends on #2 (calculations) and #3 (populated data) to have anything to display

7. Visualization & Reporting Lead

- Builds the pie chart (e.g. Chart.js) summarizing the totals
- Writes the 1000-word report: what tools were used to build the app, implications of the outputs, and controls the company should adopt to optimise operations
- Can start the report early since it's largely independent of the code being finished

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Suggested sequencing

1. **Now:** #2 (data model) and #3 (data entry) start in parallel — everything else depends on these
2. **Once #2 has a first draft:** #4 and #6 can start building against it
3. **Once #4 has a form:** #5 layers in validation/feedback
4. **Once #3 + #2 are in:** #6 and #7 can build dashboard + chart against real data
5. **#1 integrates continuously**, not just at the end
6. **#7's report** can be drafted in parallel throughout, finalized once the app is working